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Going at the heart of social cognition: is there a role for interoception in self-other distinction?

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Interoception describes the processing and awareness of bodily signals arising from visceral organs, essential for the organism's homeostatic needs. Beyond homeostasis, the integration of exteroceptive and interoceptive signals is required for the coherence of bodily self-awareness. Here, we suggest that interoception also plays a critical role in social cognition. Relating to others as individuals who are distinct from one's self requires the simultaneous yet distinct co-representation of self and others. We propose that interoceptive awareness appears to stabilise the mental representation of one's self as distinct from others. A more nuanced understanding of the role of interoception in the representation of others in relation to ourselves is vital to determine its importance in social cognition.

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Introduction

Interoception describes the afferent processing of internal bodily signals that arise from visceral organs, used to determine the physiological condition of the body and its homeostatic needs [1,2]. This definition has been broadened recently to include pleasant touch, pain and skin temperature, which are transmitted to the brain via the same spinothalamic tract and activate a similar cortical network [1,3]. There has been a long standing interest in the role of interoception in emotion processing motivated by the hypothesis that the physiological condition of the body acts as the basic substrate for feeling states and emotions [4,5]. Indeed, interoceptive signals have also been shown to modulate emotion processing; for example fearful faces are rated as more intense when presented at a particular point in the cardiac cycle [6]. In addition, functional neuroimaging studies have highlighted a substantial overlap between the neural substrates of emotion and interoception (for a review see [7]).

Beyond non-conscious interoceptive signalling, we are also capable of becoming aware of interoceptive states and changes thereof. Recent research has focused on individual differences in people's ability to access interoceptive information and highlighted how this predicts individual emotionality for similar levels of autonomic reactivity [8–11]. However, the definition of this interoceptive awareness (IAw) is very diverse within the literature and includes multiple dissociable facets. A recent review by Khalsa *et al.* [12•] elegantly summarises these different dimensions. In this paper we define IAw as representing an individual's ability to accurately and consciously perceive changes in their interoceptive state. This is dissociable from interoceptive metacognitive awareness as described by Garfinkel *et al.* [13••], which describes how good we think we are at detecting interoceptive signals; but, is comparable to previously described measures of interoceptive accuracy (IAcc), which is defined by behavioural performance on a task requiring explicit detection of interoceptive signals (e.g. the most commonly used heartbeat counting task). We therefore use the terms IAcc and IAw interchangeably throughout this paper.

Beyond emotion: the embodied foundations of self-awareness

The ubiquitous role of the insula cortex in a wide range of interoceptive, affective and social tasks [3,14,15] led to the hypothesis that the insula plays an essential functional role in body-ownership and self-awareness. Somatoparaphrenia (a loss of experienced ownership over ones limbs) is associated with right posterior insula function [16] and body ownership in the rubber hand illusion (RHI) has been associated with right mid-posterior insula cortex activity ([17], see also [18]). Interoceptive information is thought to be integrated with exteroceptive signals along the insula cortex from posterior to anterior generating a representation of self that may be integral for a wide range of socio-cognitive processes.

Recent work has sought to understand not simply *where* in the brain but *how* exteroceptive and interoceptive signals are integrated to produce a unified experience of the bodily self by determining the relationship between IAcc

and body-ownership. The RHI and the Enfacement Illusion are used to modulate body-ownership in response to exteroceptive multisensory stimulation and reveal the malleability of an individual's representation of self; this is driven by the brain's integration of multisensory exteroceptive signals to update the mental [19] and physiological [20] representation of one's body [21]. Tsakiris *et al.* [22] showed that a stronger RHI, measured behaviourally (i.e. increased proprioceptive drift) and homeostatically (i.e. a drop in skin temperature), was induced in those with lower compared to higher IAcc (see also [23]). Similar negative correlations between the effects of multisensory stimulation and trait-like levels of IAcc were observed in the Enfacement Illusion: participants with lower IAcc experienced a stronger sense of identification with another person's face [24]. These findings suggest that those with higher IAcc have a stronger sense of self, which is grounded in their interoceptive states, making them less susceptible to body illusions. If exteroceptive influences highlight the malleability of body-awareness, awareness of interoceptive signals seem to serve the stability of the self, reflecting a psychological consequence of the biologically necessary function of homeostasis.

Beyond these correlational observations, other studies used cardiac signals as inducers of changes in body-ownership. The substitution of exteroceptive visuo-tactile stimulation with cardio-visual stimulation (i.e. a combination of interoceptive and exteroceptive signals) led to the study of the potentially causal role of interoceptive signals for body-awareness. Suzuki *et al.* (2013) showed that looking at a virtual hand that pulsated in synchrony with one's heartbeat led to changes in body-ownership similar to those reported in the classic RHI (see also [25,26]). Here enhancing the saliency of interoceptive signals modulated perceived body-ownership and self-identification [27,28••], especially for people with higher levels of IAcc [26,29].

Beyond selfhood: interoception for alteroception

The question of how bodily self-awareness emerges in the brain is intrinsically linked to the question of how we relate to others. It is through intersubjective interactions in early life that the self emerges [30–32]. This ever-present social relatedness has a dual nature: on one hand the self is built and represented as distinct from others, but on the other hand the boundaries between self and other are not fixed. Predictive accounts of mentalising suggest that we generate hypotheses about other's mental and bodily states based on our own archive of behaviours, which are used to predict other's behaviour. Recently this has been extended to include interoceptive states [33•,34]. At the ontogenetic level carer–infant bodily interactions enable infants to develop a predictive model of their own interoception [35•]. Using this we can then integrate exteroceptive input and learned contextual knowledge to infer another person's interoceptive state [33•]. A fundamental question in social neuroscience is how it becomes possible to distinguish between self and other at the very time that we are trying to relate to each other? For example, how can I share another individual's pain without forgetting it is not *my* pain?

Emotional contagion, mimicry, body resonance, perspective-taking, theory of mind (ToM) and egocentricity biases have been used to operationalize different facets of empathy. Although its exact meaning is still debated, empathy is considered one of the hallmarks of social cognition [36,37]. A critical but unresolved issue is this question of 'self-other' overlap [38]. 'Self-other' overlap is thought to arise when an observer engages in an isomorphic state (e.g. same emotion) to the person observed. However, what is or should be the extent of this overlap? Interoception may play a key role in enabling us to navigate the different degrees of social relatedness, by allowing us to correctly attribute the origin of bodily and mental states to oneself or to others.

Previous theories of self-other distinction focus on the central role of the sensorimotor system in generating a sense of agency and body-ownership through the integration of proprioceptive and exteroceptive inputs [39–41]. Our self-awareness comes from the understanding of how we interact with the environment from a first person perspective. Indeed, the default *modus operandi* of the social brain seems to be to represent one's own self (e.g. one's own perspective, emotion, beliefs), therefore switching from self to other, to achieve a partial co-representation of self and other, is an effortful process that to some extent requires the attenuation of self-representations [42]. Indeed, inhibitory control is deemed a necessary component of mentalising [41] and inhibition of imitative behaviour is used as a measure of online control of self-other representations [43,44]. We extend these ideas to incorporate interoception.

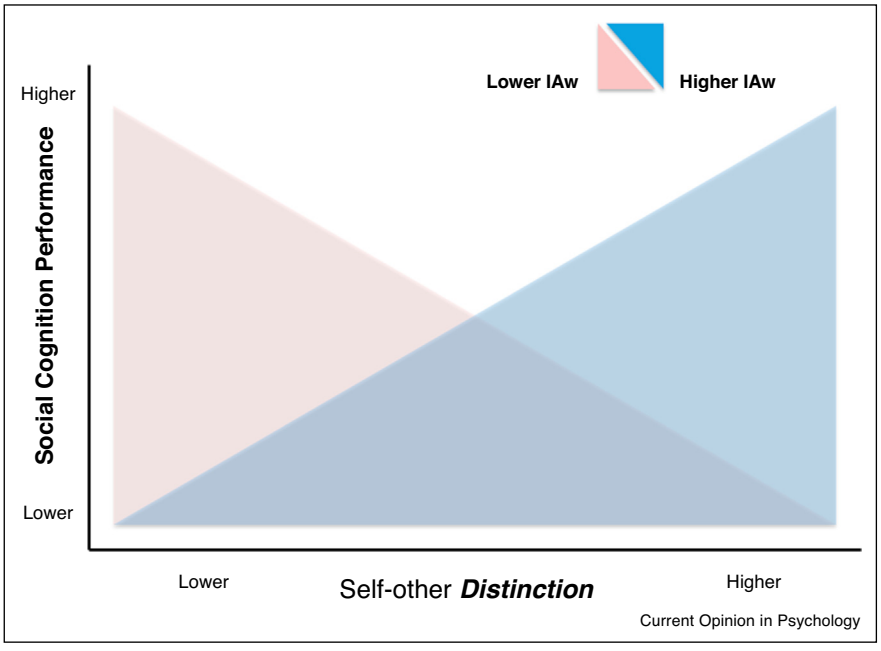
We propose that interoceptive processing acts to stabilise the model of our self, such that we can readily attribute emotional and mental states to the self or to others without blurring the distinction between 'self' and 'other'. In line with recent theoretical accounts of self-processing, we characterise the underlying mechanism for this within the Predictive Coding framework [21,27,45]. Probabilistic representations of the self are generated by the integration of top-down 'predictions' about the bodily state and bottom-up 'prediction errors' (PEs) across interoceptive and exteroceptive modalities. The importance or salience of these inputs and relative contribution to the resultant percept is determined by their 'precision-weighting' or inverse variance. We hypothesise that stability in our model of the self comes from the level of precision attributed to interoceptive relative to

exteroceptive PEs, which is optimised by attention [34,46••]. When integrating a precise interoceptive PE with a relatively less precise exteroceptive PE, the interoceptive input will dominate the resultant percept and optimise higher level beliefs about the self. Precise interoception throughout development will therefore accommodate the experience of the self as stable and continuous.

The importance of this can be demonstrated with the RHI [47]. The experience of the illusion depends on how PEs across modalities are explained away. Reducing the precision of interoceptive PEs will increase the relative precision of exteroceptive PEs and update higher-order beliefs about the ownership of the rubber hand [26,48]. We hypothesise that individuals with lower IAw will be more readily able to attenuate relatively less precise interoceptive PE signals, therefore high level beliefs about the self are easily modified to explain away more precise exteroceptive PEs. Alternatively, those with higher IAw will have more precise interoceptive PEs, which are not as easily attenuated relative to conflicting exteroceptive PEs. In line with this theory, those with high IAcc are less susceptible to the RHI [22] and the Enfacement Illusion [24].

Based on these ideas we can generate alternative hypotheses regarding the impact of IAw on social-cognitive processes with a different magnitude of self-other overlap. Emotional contagion, the most basic level of empathy, is the process by which our own interoceptive state automatically mimics that of the observed person, and is associated with activity across the network of neural areas activated when the person experiences the emotion themselves [49]. We hypothesise that those with lower IAw may be more readily able to attenuate less precise interoceptive PEs in order to update higher order beliefs about another person's emotional state. At a higher level of empathy, an individual may be required to represent another's emotional state that may be incongruent with their own emotional state. Consider a couple arguing: the spouse is angry, her blood pressure and heart rate increase as she quarrels with her husband who suddenly starts crying eliciting conflicting interoceptive and emotional reactions in the wife. The co-presence of her own-generated interoceptive states and those generated by her husband's reaction are now conflicting but co-represented both cognitively and physiologically. In this context, individual differences in the relative precision-weighting of interoceptive and expteroceptive states may be important for maintaining a self-other distinction, therefore higher IAw may be advantageous (see Figure 1). Understanding others may therefore require at least a 'good enough' representation of one's own (interoceptive) state because the key element in representing other's states is how ***their*** states affect ***us***.

Figure 1



illustrates a nuanced perspective of how interoceptive awareness (IAw) may impact on social cognitive performance depending on the levels of self-other overlap in the task. For tasks with a low level of self-other distinction, for example emotional contagion, it may be advantageous to have lower level of IAw, and we predict that such individuals will display greater emotional contagion. In contrast, when a social task requires a high level of self-other distinction, individuals with higher IAw may be better able to understand the emotional state of others because these individuals have a more stable representation of their own bodily self, which prevents the blurring of self and other resulting in improved performance.

The proposed interaction between levels of IAw and 'self-other' overlap, which is integral to all social interactions, can be empirically tested by adopting a hierarchical approach: firstly, emotion contagion where the self-other overlap is almost complete, and demands on self-other distinction are minimal; secondly, empathy for pain where there is partial self-other sharing; and thirdly, a complete distinction between self and other where individuals must relate to each other in an empathetic way whilst maintaining a dissociable emotional and interoceptive state, which can be measured using an egocentricity bias task [50]. Individual differences in trait interoception measures can then be compared to social cognition measures at each level of self-other overlap to determine their inter-relationship.

Evidence and future directions for the role of interoception in social cognition

Empathy is a multidimensional construct that necessitates a more fine-grained, nuanced approach to understanding the influence of interoception on empathy. While the links between interoceptive processing and the nuanced examples of empathy remain largely unexplored, there is some supporting evidence for the hypothesis we put forward. Individuals with Autism Spectrum Disorder (ASD), characterised by deficits in social cognition, have impaired interoceptive processing [51,52]. However, their interoceptive deficits appear to be mainly linked to the co-morbid presence of alexithymia (a difficulty identifying emotions) rather than mentalising ability [53]. In neurotypical participants, IAcc has been shown to correlate with performance on emotional items in the Movie for the Assessment of Social Cognition (MASC), but not for non-emotional items [54••], and with empathy for pain [55]. However, no association was found between IAcc and cognitive or self-report measures of empathy [56] but IAcc did predict altruistic behaviour [57]. Interoception may be more relevant for those emotional aspects of social cognition that influence our ability to empathise with others.

The field should also seek to go beyond correlational designs and trait measures of interoception. The validity of the current tasks used to measure trait IAcc and IAw has been questioned [58,59], therefore novel tasks to measure state and trait interoception are required. Presenting stimuli at specific points in the cardiac cycle allows control over the autonomic context with which stimuli are perceived. At systole, during a heartbeat, arterial baroreceptors signal to the brain, but at diastole, between heartbeats, this pathway is silent. Baroreceptor firing during systole amplifies threat processing to fearful faces [6], decreases memory for words [60] and increases the expression of threat-related racial stereotyping [61]. Arousal can also be modulated using masked stimuli to induce general increases in autonomic activity [62] and the effect of this on socio-cognitive processing can be explored. Modulations in interoceptive modalities outside of the cardiovascular system are harder to control, however other modalities, such as the gastrointestinal system [63] and respiratory system [64,65] are now being studied to determine the specificity and contribution of different interoceptive inputs to social cognition. Importantly, direct and controlled manipulations of interoceptive signalling, such as the use of CO_2 inhalation, beta blockers or water load, will enable us to probe the causal impact of interoception on tasks requiring self-other processing. Moreover, the ontogenetic origins of interoception [66] must also be explored. Using these methods we can better elucidate the causal role of interoception and awareness thereof on the processing of self-related and socially relevant stimuli.

Conclusion

Interoception plays an important role in the mental representation of the self [35•]. Those with lower interoceptive awareness are more susceptible to bodily illusions due to a greater malleability of their self-representation. This has important consequences for social cognition and in particular for our ability to empathise. One needs an accurate sense of her own interoceptive body in order to be able to correctly attribute mental and bodily states to oneself or another person and understand the causes of those states, whilst maintaining the distinction between self and other. Empirical evidence using paradigms that manipulate the interoceptive state are necessary to determine the nuanced relationship between interoceptive processing and the self-other overlap, which is integral to understanding the role of interoception in social cognition.

Conflict of interest statement

Nothing declared.

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References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. Craig AD: **How do you feel? Interoception: the sense of the physiological condition of the body**. *Nat Rev Neurosci* 2002, **3**:655-666 http://dx.doi.org/10.1038/nrn894.

2. Cameron O: *Visceral Sensory Neuroscience: Interoception*. Oxford: Oxford University Press; 2002.

3. Craig AD: **How do you feel — now? The anterior insula and human awareness**. *Nat Rev Neurosci* 2009, **10**:59-70 http://dx.doi.org/10.1038/nrn2555.

4. Cannon WB: **The James–Lange theory of emotions: a critical examination and an alternative theory. By Walter B. Cannon, 1927**. *Am J Psychol* 1987, **100**:567-586 http://dx.doi.org/10.2307/1415404.

5. James W: **What is an emotion?** *Mind* 1884, **9**:188-205 http://dx.doi.org/10.1093/mind/LI.202.200.

6. Garfinkel SN, Minati L, Gray MA, Seth AK, Dolan RJ, Critchley HD: **Fear from the heart: sensitivity to fear stimuli depends on individual heartbeats**. *J Neurosci* 2014, **34**:6573-6582 http://dx.doi.org/10.1523/JNEUROSCI.3507-13.2014.

7. Critchley HD, Garfinkel SN: **Interoception and emotion**. *Curr Opin Psychol* 2017, **17**:7-14 http://dx.doi.org/10.1016/j.copsyc.2017.04.020.

8. Wiens S, Mezzacappa ES, Katkin ES: **Heartbeat detection and the experience of emotions**. *Cogn Emot* 2000, **14**:417-427 http://dx.doi.org/10.1080/026999300378905.

9. Wiens S: **Interoception in emotional experience**. *Curr Opin Neurol* 2005, **18** http://w3.psychology.su.se/staff/sws/Wiens05_interoception.pdf (accessed 10.11.17).

10. Pollatos O, Schandry R: **Emotional processing and emotional memory are modulated by interoceptive awareness**. *Cogn Emot* 2008, **22**:272-287 http://dx.doi.org/10.1080/02699930701357535.

11. Barrett LF, Quigley KS, Bliss-Moreau E, Aronson KR: **Interoceptive sensitivity and self-reports of emotional experience**. *J Pers Soc Psychol* 2004, **87**:684-697 http://dx.doi.org/10.1037/0022-3514.87.5.684.

12. • Khalsa SS, Adolphs R, Cameron OG, Critchley HD, Davenport PW, Feinstein JS, Feusner JD, Garfinkel SN, Lane RD, Mehling WE, Meuret AE, Nemeroff CB, Oppenheimer S, Petzschner FH, Pollatos O, Rhudy JL, Schramm LP, Simmons WK, Stein MB, Stephan KE, Van den Bergh O, Van Diest I, von Leupoldt A, Paulus MP, Ainley V, Al Zoubi O, Aupperle R, Avery J, Baxter L, Benke C, Berner L, Bodurka J, Breese E, Brown T, Burrows K, Cha Y-H, Clausen A, Deville D, Duncan L, Duquette P, Ekhtiari H, Fine T, Ford B, Garcia Cordero I, Gleghorn D, Guereca Y, Harrison NA, Hassanpour M, Hechler T, Heller A, Hellman N, Herbert B, Jarrahi B, Kerr K, Kirlic N, Klabunde M, Kraynak T, Kriegsman M, Kroll J, Kuplicki R, Lapidus R, Le T, Hagen KL, Mayeli A, Morris A, Naqvi N, Oldroyd K, Pané-Farré C, Phillips R, Poppa T, Potter W, Puhl M, Safron A, Sala M, Savitz J, Saxon H, Schoenhals W, Stanwell-Smith C, Teed A, Terasawa Y, Thompson K, Toups M, Umeda S, Upshaw V, Victor T, Wierenga C, Wohlrab C, Yeh H, Yoris A, Zeidan F, Zotev V, Zucker N: **Interoception and mental health: a roadmap**. *Biol Psychiatry Cogn Neurosci Neuroimaging* 2017 http://dx.doi.org/10.1016/J.BPSC.2017.12.004.

This paper provides a useful introduction to interoception and summarises the current methodological issues facing interoception research. The authors also highlight how interoception may be dysfunctional in particular psychopathologies which is of particular relevance for the framework put forward here as many of these disorders also include symptoms of abnormal self-other processing.

13. •• Garfinkel SN, Seth AK, Barrett AB, Suzuki K, Critchley HD: **Knowing your own heart: distinguishing interoceptive accuracy from interoceptive awareness**. *Biol Psychol* 2015, **104**:65-74 http://dx.doi.org/10.1016/j.biopsycho.2014.11.004.

This seminal study demonstrated for the first time that interoception is a multi-dimensional construct and highlighted the importance of measuring interoception across different levels of subjective and objective awareness in order to understand how interoception at each of these levels affects cognition.

14. Stern ER, Grimaldi SJ, Muratore A, Murrough J, Leibu E, Fleysher L, Goodman WK, Burdick KE: **Neural correlates of interoception: effects of interoceptive focus and relationship to dimensional measures of body awareness**. *Hum Brain Mapp* 2017, **38**:6068-6082 http://dx.doi.org/10.1002/hbm.23811.

15. Schulz SM: **Neural correlates of heart-focused interoception: a functional magnetic resonance imaging meta-analysis**. *Philos Trans R Soc Lond B Biol Sci* 2016, **371**:20160018 http://dx.doi.org/10.1098/rstb.2016.0018.

16. Karnath H-O, Baier B: **Right insula for our sense of limb ownership and self-awareness of actions**. *Brain Struct Funct* 2010, **214**:411-417 http://dx.doi.org/10.1007/s00429-010-0250-4.

17. Tsakiris M, Hesse MD, Boy C, Haggard P, Fink GR: **Neural signatures of body ownership: a sensory network for bodily self-consciousness**. *Cereb Cortex* 2007, **17**:2235-2244 http://dx.doi.org/10.1093/cercor/bhl131.

18. Guterstam A, Abdulkarim Z, Ehrsson HH: **Illusory ownership of an invisible body reduces autonomic and subjective social anxiety responses**. *Sci Rep* 2015, **5**:9831 http://dx.doi.org/10.1038/srep09831.

19. Longo MR, Schüür F, Kammers MPM, Tsakiris M, Haggard P: **Self awareness and the body image**. *Acta Psychol (Amst)* 2009, **132**:166-172 http://dx.doi.org/10.1016/j.actpsy.2009.02.003.

20. Moseley GL, Olthof N, Venema A, Don S, Wijers M, Gallace A, Spence C: **Psychologically induced cooling of a specific body part caused by the illusory ownership of an artificial counterpart**. *Proc Natl Acad Sci U S A* 2008, **105**:13169-13173 http://dx.doi.org/10.1073/pnas.0803768105.

21. Apps MAJ, Tsakiris M: **The free-energy self: a predictive coding account of self-recognition**. *Neurosci Biobehav Rev* 2014, **41**:85-97 http://dx.doi.org/10.1016/j.neubiorev.2013.01.029.

22. Tsakiris M, Tajadura-Jiménez A, Costantini M: **Just a heartbeat away from one's body: interoceptive sensitivity predicts malleability of body-representations**. *Proc Biol Sci* 2011, **278**:2470-2476 http://dx.doi.org/10.1098/rspb.2010.2547.

23. Schauder KB, Mash LE, Bryant LK, Cascio CJ: **Interoceptive ability and body awareness in autism spectrum disorder**. *J Exp Child Psychol* 2015, **131**:193-200 http://dx.doi.org/10.1016/j.jecp.2014.11.002.

24. Tajadura-Jiménez A, Tsakiris M: **Balancing the "inner" and the "outer" self: interoceptive sensitivity modulates self-other boundaries**. *J Exp Psychol Gen* 2014, **143**:736-744 http://dx.doi.org/10.1037/a0033171.

25. Aspell JE, Heydrich L, Marillier G, Lavanchy T, Herbelin B, Blanke O: **Turning body and self inside out: visualized heartbeats alter bodily self-consciousness and tactile perception**. *Psychol Sci* 2013, **24**:2445-2453 http://dx.doi.org/10.1177/0956797613498395.

26. Sel A, Azevedo RT, Tsakiris M: **Heartfelt self: cardio-visual integration affects self-face recognition and interoceptive cortical processing**. *Cereb Cortex* 2016 http://dx.doi.org/10.1093/cercor/bhw296.

27. Seth AK: **Interoceptive inference, emotion, and the embodied self**. *Trends Cogn Sci* 2013, **17**:565-573 http://dx.doi.org/10.1016/J.TICS.2013.09.007.

28. •• Tsakiris M: **The multisensory basis of the self: from body to identity to others**. *Q J Exp Psychol* 2017, **70**:597-609 http://dx.doi.org/10.1080/17470218.2016.1181768.

This recent review provides a comprehensive overlook of recent advances in the study of body-ownership and discusses the role of the exteroceptive and interoceptive signals for body-awareness and an understanding of the self for social relationships.

29. Suzuki K, Garfinkel SN, Critchley HD, Seth AK: **Multisensory integration across exteroceptive and interoceptive domains modulates self-experience in the rubber-hand illusion**. *Neuropsychologia* 2013, **51**:2909-2917 http://dx.doi.org/10.1016/j.neuropsychologia.2013.08.014.

30. Reddy V: **On being the object of attention: implications for self-other consciousness**. *Trends Cogn Sci* 2003, **7**:397-402 http://dx.doi.org/10.1016/S1364-6613(03)00191-8.

31. Rochat P: **Five levels of self-awareness as they unfold early in life**. *Conscious Cogn* 2003, **12**:717-731 http://dx.doi.org/10.1016/S1053-8100(03)00081-3.

32. Schilbach L, Timmermans B, Reddy V, Costall A, Bente G, Schlicht T, Vogeley K: **Toward a second-person neuroscience**. *Behav Brain Sci* 2013, **36**:393-414 http://dx.doi.org/10.1017/S0140525X12000660.

33. • Ondobaka S, Kilner J, Friston K: **The role of interoceptive inference in theory of mind**. *Brain Cogn* 2017, **112**:64-68 http://dx.doi.org/10.1016/j.bandc.2015.08.002.

This paper highlights how inferring others internal states can be used to understand other people's mental states and explain the underlying causes of particular behaviours. Interoceptive inference is combined into a Bayesian predictive coding framework for perception and action.

34. Seth AK, Friston KJ: **Active interoceptive inference and the emotional brain**. *Philos Trans R Soc Lond B Biol Sci* 2016, **371**:20160007 http://dx.doi.org/10.1098/rstb.2016.0007.

35. • Fotopoulou A, Tsakiris M: **Mentalizing homeostasis: the social origins of interoceptive inference**. *Neuropsychoanalysis* 2017, **19**:3-28 http://dx.doi.org/10.1080/15294145.2017.1294031.
This article discusses the importance of social interactions throughout development in generating a model of how our homeostatic regulation is altered in different contexts and therefore prove essential in creating the representation of our embodied self.

36. Bernhardt BC, Singer T: **The neural basis of empathy**. *Annu Rev Neurosci* 2012, **35**:1-23 http://dx.doi.org/10.1146/annurev-neuro-062111-150536.

37. Decety J: **The neuroevolution of empathy**. *Ann N Y Acad Sci* 2011, **1231**:35-45 http://dx.doi.org/10.1111/j.1749-6632.2011.06027.x.

38. Preston SD, Hofelich AJ: **The many faces of empathy: parsing empathic phenomena through a proximate, dynamic-systems view of representing the other in the self**. *Emot Rev* 2012, **4**:24-33 http://dx.doi.org/10.1177/1754073911421378.

39. Jeannerod M, Pacherie E: **Agency, simulation and self-identification**. *Mind Lang* 2004, **19**:113-146 http://dx.doi.org/10.1111/j.1468-0017.2004.00251.x.

40. Blakemore S-J, Decety J: **From the perception of action to the understanding of intention**. *Nat Rev Neurosci* 2001, **2**:529 http://dx.doi.org/10.1038/35086023.

41. Decety J, Sommerville JA: **Shared representations between self and other: a social cognitive neuroscience view**. *Trends Cogn Sci* 2003, **7**:527-533 http://dx.doi.org/10.1016/J.TICS.2003.10.004.

42. Bird G, Viding E: **The self to other model of empathy: providing a new framework for understanding empathy impairments in psychopathy, autism, and alexithymia**. *Neurosci Biobehav Rev* 2014, **47**:520-532 http://dx.doi.org/10.1016/j.neubiorev.2014.09.021.

43. Santiesteban I, Banissy MJ, Catmur C, Bird G: **Enhancing social ability by stimulating right temporoparietal junction**. *Curr Biol* 2012, **22**:2274-2277 http://dx.doi.org/10.1016/j.cub.2012.10.018.

44. Brass M, Heyes C: **Imitation: is cognitive neuroscience solving the correspondence problem?** *Trends Cogn Sci* 2005, **9**:489-495 http://dx.doi.org/10.1016/J.TICS.2005.08.007.

45. Sel A: **Predictive codes of interoception, emotion, and the self**. *Front Psychol* 2014, **5**:189 http://dx.doi.org/10.3389/fpsyg.2014.00189.

46. •• Ainley V, Apps MAJ, Fotopoulou A, Tsakiris M: **"Bodily precision": a predictive coding account of individual differences in interoceptive accuracy**. *Philos Trans R Soc B Biol Sci* 2016, **371**:20160003 http://dx.doi.org/10.1098/rstb.2016.0003.
This theoretical perspective describes how differences in IAcc can be explained by differences in the optimisation of precision between and within modalities. It is this idea which is extended in the current paper to explain self-other processing.

47. Tsakiris M, Haggard P: **The rubber hand illusion revisited: visuotactile integration and self-attribution**. *J Exp Psychol Hum Percept Perform* 2005, **31**:80-91 http://dx.doi.org/10.1037/0096-1523.31.1.80.

48. Zeller D, Friston KJ, Classen J: **Dynamic causal modeling of touch-evoked potentials in the rubber hand illusion**. *Neuroimage* 2016, **138**:266-273 http://dx.doi.org/10.1016/J.NEUROIMAGE.2016.05.065.

49. Preston SD, de Waal FBM: **Empathy: its ultimate and proximate bases**. *Behav Brain Sci* 2002, **25** 1-20-71. http://www.ncbi.nlm.nih.gov/pubmed/12625087 (accessed 12.11.17).

50. Silani G, Lamm C, Ruff CC, Singer T: **Right supramarginal gyrus is crucial to overcome emotional egocentricity bias in social judgments**. *J Neurosci* 2013, **33**:15466-15476 http://dx.doi.org/10.1523/JNEUROSCI.1488-13.2013.

51. Garfinkel SN, Tiley C, O'Keeffe S, Harrison NA, Seth AK, Critchley HD: **Discrepancies between dimensions of interoception in autism: Implications for emotion and anxiety**. *Biol Psychol* 2016, **114**:117-126 http://dx.doi.org/10.1016/j.biopsycho.2015.12.003.

52. Noel J-P, Lytle M, Cascio C, Wallace MT: **Disrupted integration of exteroceptive and interoceptive signaling in autism spectrum disorder**. *Autism Res* 2017 http://dx.doi.org/10.1002/aur.1880.

53. Shah P, Hall R, Catmur C, Bird G: **Alexithymia, not autism, is associated with impaired interoception**. *Cortex* 2016, **81**:215-220 http://dx.doi.org/10.1016/j.cortex.2016.03.021.

54. •• Shah P, Catmur C, Bird G: **From heart to mind: linking interoception, emotion, and theory of mind**. *Cortex* 2017, **93**:220-223 http://dx.doi.org/10.1016/j.cortex.2017.02.010.
This study found that IAcc only predicted performance on emotional items in a Theory of Mind task not non-emotional items. This suggests that good IAcc is only advantageous in social contexts when emotion understanding is required.

55. Grynberg D, Pollatos O: **Perceiving one's body shapes empathy**. *Physiol Behav* 2015, **140**:54-60 http://dx.doi.org/10.1016/j.physbeh.2014.12.026.

56. Ainley V, Maister L, Tsakiris M: **Heartfelt empathy? No association between interoceptive awareness, questionnaire measures of empathy, reading the mind in the eyes task or the director task**. *Front Psychol* 2015, **6**:554 http://dx.doi.org/10.3389/fpsyg.2015.00554.

57. Piech RM, Strelchuk D, Knights J, Hjälmheden JV, Olofsson JK, Aspell JE: **People with higher interoceptive sensitivity are more altruistic, but improving interoception does not increase altruism**. *Sci Rep* 2017, **7**:15652 http://dx.doi.org/10.1038/s41598-017-14318-8.

58. Brener J, Ring C: **Towards a psychophysics of interoceptive processes: the measurement of heartbeat detection**. *Philos Trans R Soc B Biol Sci* 2016, **371**:20160015 http://dx.doi.org/10.1098/rstb.2016.0015.

59. Ring C, Brener J, Knapp K, Mailloux J: **Effects of heartbeat feedback on beliefs about heart rate and heartbeat counting: a cautionary tale about interoceptive awareness**. *Biol Psychol* 2015, **104**:193-198 http://dx.doi.org/10.1016/j.biopsycho.2014.12.010.

60. Garfinkel SN, Barrett AB, Minati L, Dolan RJ, Seth AK, Critchley HD: **What the heart forgets: cardiac timing influences memory for words and is modulated by metacognition and interoceptive sensitivity**. *Psychophysiology* 2013, **50**:505-512 http://dx.doi.org/10.1111/psyp.12039.

61. Azevedo RT, Garfinkel SN, Critchley HD, Tsakiris M: **Cardiac afferent activity modulates the expression of racial stereotypes**. *Nat Commun* 2017, **8**:13854 http://dx.doi.org/10.1038/ncomms13854.

62. Allen M, Frank D, Schwarzkopf DS, Fardo F, Winston JS, Hauser TU, Rees G: **Unexpected arousal modulates the influence of sensory noise on confidence**. *Elife* 2016, **5** http://dx.doi.org/10.7554/eLife.18103.

63. Richter CG, Babo-Rebelo M, Schwartz D, Tallon-Baudry C: **Phase-amplitude coupling at the organism level: the amplitude of spontaneous alpha rhythm fluctuations varies with the phase of the infra-slow gastric basal rhythm**. *Neuroimage* 2017, **146**:951-958 http://dx.doi.org/10.1016/j.neuroimage.2016.08.043.

64. Garfinkel SN, Manassei MF, Hamilton-Fletcher G, In den Bosch Y, Critchley HD, Engels M: **Interoceptive dimensions across cardiac and respiratory axes**. *Philos Trans R Soc Lond B Biol Sci* 2016, **371**:20160014 http://dx.doi.org/10.1098/rstb.2016.0014.

65. Murphy J, Catmur C, Bird G: **Alexithymia is associated with a multidomain, multidimensional failure of interoception: evidence from novel tests**. *J Exp Psychol Gen* 2017 http://dx.doi.org/10.1037/xge0000366.

66. Maister L, Tang T, Tsakiris M: **Neurobehavioral evidence of interoceptive sensitivity in early infancy**. *Elife* 2017, **6** http://dx.doi.org/10.7554/eLife.25318.